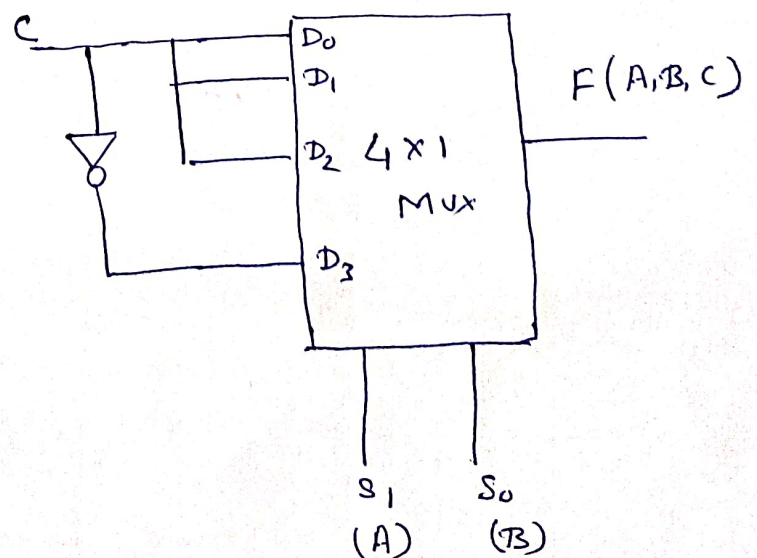


Q.1 a) Implement the following function with a MUX
 $F(A, B, C) = \sum m(1, 3, 5, 6)$
 choose A and B as select inputs

Ans:- The truth table for F and the implementation of $f(A, B, C) = \sum m(1, 3, 5, 6)$ using a 4:1 MUX with select input A and B

$S_1 \quad S_0$			
A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

A	B	F
0	0	C
0	1	C
1	0	C
1	1	$\bar{C}$



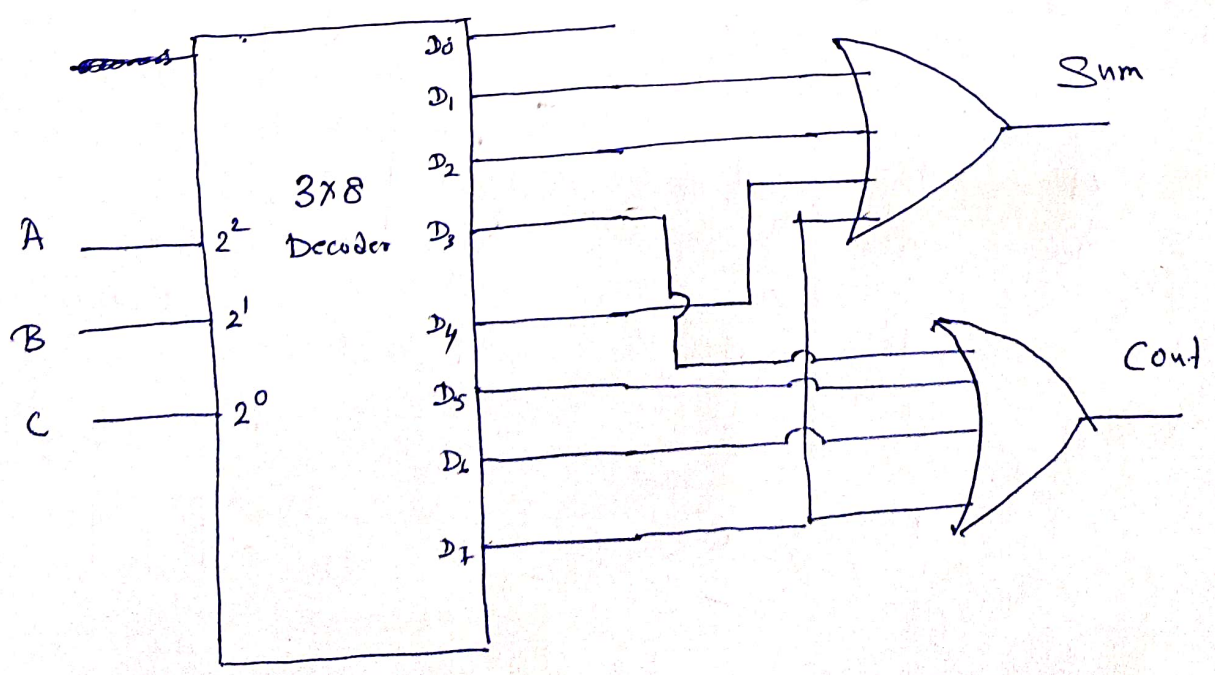
Q.6 Design Full adder using Suitable decoder and logic gates

Ans:- Truth table for Full adder

A	B	C	Sum	Cont	
0	0	0	0	0	→ D ₀
0	0	1	1	0	→ D ₁
0	1	0	1	0	→ D ₂
0	1	1	0	1	→ D ₃
1	0	0	1	0	→ D ₄
1	0	1	0	1	→ D ₅
1	1	0	0	1	→ D ₆
1	1	1	1	1	→ D ₇

$$\text{Sum} = A \oplus B \oplus C_{in} = \sum m(1, 2, 4, 7) = D_1 + D_2 + D_4 + D_7$$

$$\text{Cont / Carry} = AB + (A \oplus B)C_{in} = \sum m(3, 5, 6, 7) = D_3 + D_5 + D_6 + D_7$$



Q2(a)

$$F = (A \oplus B) \oplus (AB)$$

$$= (\overline{A \oplus B}) \cdot AB + (A \oplus B) \overline{AB}$$

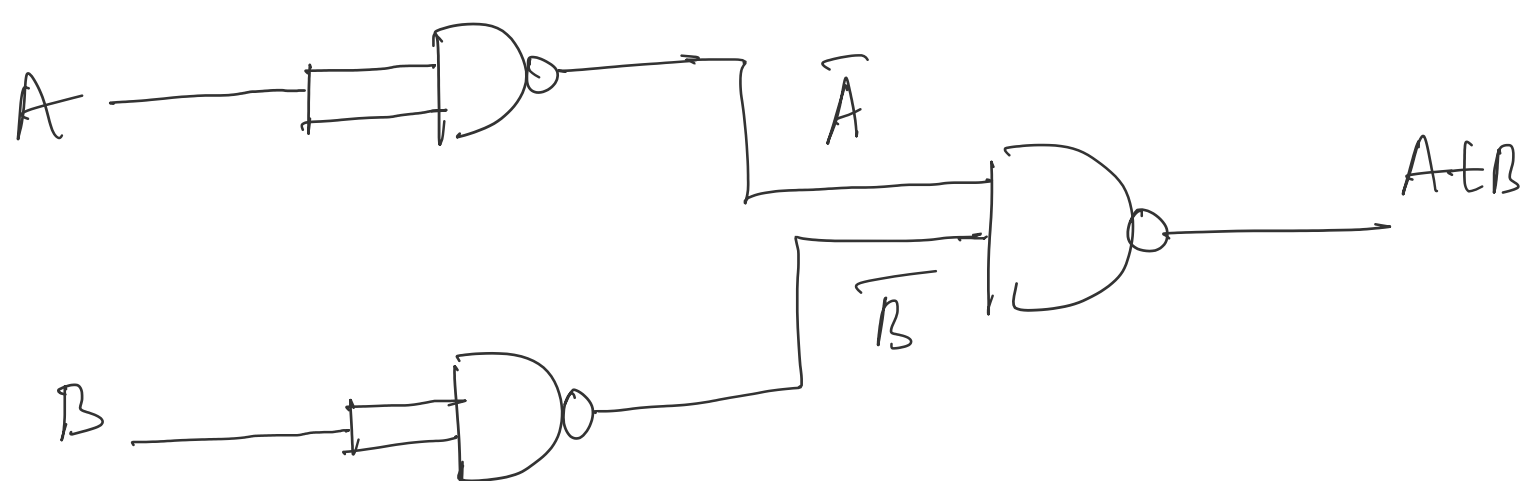
$$= (\overline{AB} + AB) AB + (A\overline{B} + \overline{A}B)(\overline{A} + \overline{B})$$

$$= AB + A\overline{B} + \overline{A}B$$

$$= A(B + \overline{B}) + \overline{A}B$$

$$= A + \overline{A}B = (A + \overline{A})(A + B)$$

$$= A + B$$

(b)Truth table

P	N	Q
0	0	0
0	1	NC
1	0	Q
1	1	1

Excitation table

Q_n	Q_{n+1}	P	N
0	0	0	X
0	1	1	X
1	0	X	0
1	1	X	1

Derivation of state equation

Q_n	P	N	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Q_n	P \ N	00	01	11	10
0				1	1
1			1	1	

$\overline{Q_n}P$

Q_nN

$$Q_{n+1} = \overline{Q_n}P + Q_nN$$

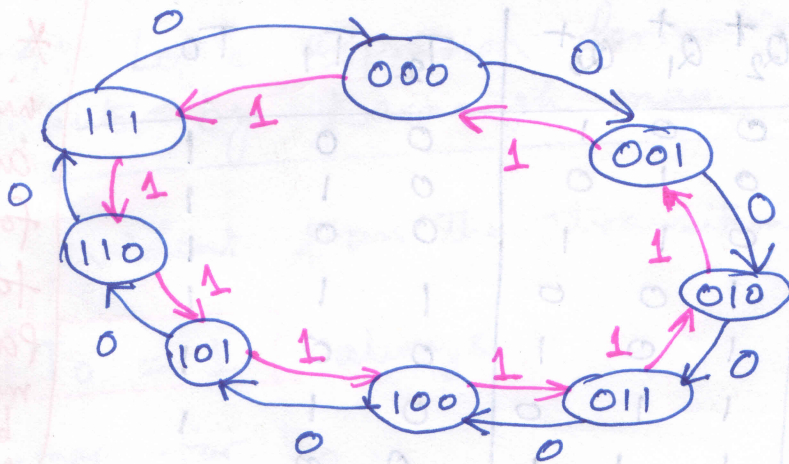
Solution

Q3. Design a 3-bit synchronous binary Up/Down counter using T flip-flops with detailed steps including state diagram, next-state table, flip-flop transition table and implementation using logic gates and flip-flops.

Ans:

Step 1: State Diagram

The first step in the design of a counter is to create a state diagram. State diagrams for Up/Down counting is shown below:



$M=0$: Up Counting

$M=1$: Down Counting

* 2 marks will be awarded for showing both Up and Down counting, else partial marking will be given. only Up counting/Down counting -

1 marks

Step 2: Next-state Table

Once the state diagram is defined, the second step is to derive a next-state table which lists each state of counter along with the corresponding next state.

Step 3: Flip-Flop Transition Table

Once next state table is defined, the inputs to the T flip-flops can be defined using flip-flop transition table. Both the next-state table and flip-flop transition table are combined in the table shown below.

M	Q ₂	Q ₁	Q ₀	Q ₂ ⁺	Q ₁ ⁺	Q ₀ ⁺	T ₂	T ₁	T ₀
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	1	1	0	0	0	1

* 5 marks will be awarded for this table. Partial marks will be awarded otherwise.

* only transition table w/o next-state table - 3 marks

Corresponding excitation table of T flip-flop is :

Output Transition		Flip-flop inputs
Q	Q^+	T
0	0	0
0	1	1
1	0	1
1	1	0

Here, Q_2, Q_1, Q_0 are present states; Q_2^+, Q_1^+, Q_0^+ are next states; and T_2, T_1 and T_0 are the inputs to the T flip-flops.

Step 4: Logic expression derivation for flip-flop inputs by Karnaugh map minimization

As evident from the transition table in step 3,

$T_0 = 1$ always.

K-map for T_2 :

Q_1Q_0				
MQ_2	00	01	11	10
00			1	
01			1	
11	1			
10	1			

$$T_2 = Q_1 Q_0 \bar{M} + \bar{Q}_1 \bar{Q}_0 M$$

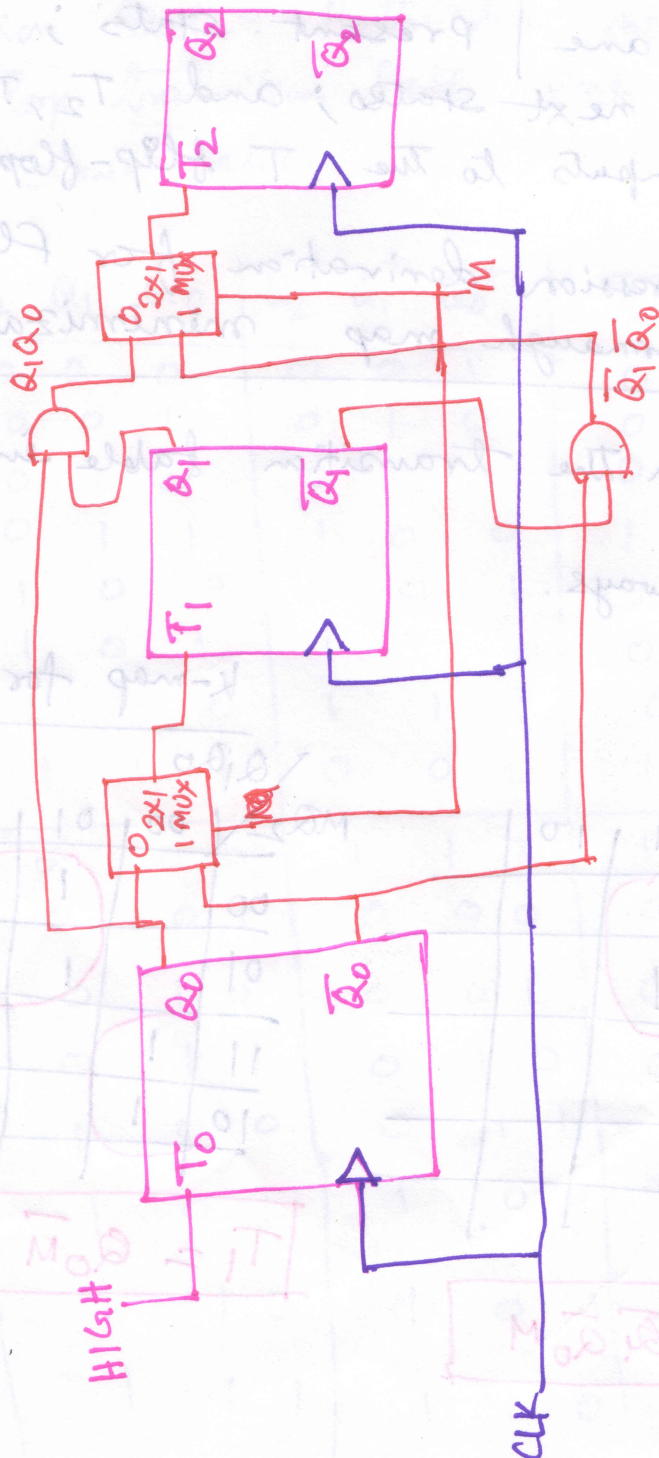
K-map for T_1

Q_1Q_0				
MQ_2	00	01	11	10
00		1	1	
01		1	1	
11	1			1
10	1			1

$$T_1 = Q_0 \bar{M} + \bar{Q}_0 M$$

Step 5: Implementation of Counter

The final step is to implement the combinational logic from the expressions for the T flip-flop inputs and connect the flip-flops to form the complete 3-bit synchronous Up/Down Counter as shown in the figure below:



* Instead of 2x1 MUX, the logic can be implemented using AND, OR and NOT gates also.

3 marks will be awarded if both K-map and implementation is shown, else partial marks will be awarded

Only K-map - $1\frac{1}{2}$ marks
only implementation - $1\frac{1}{2}$ marks

$$M_0 \bar{Q}_0 + \bar{M}_0 Q_0 = T_0$$

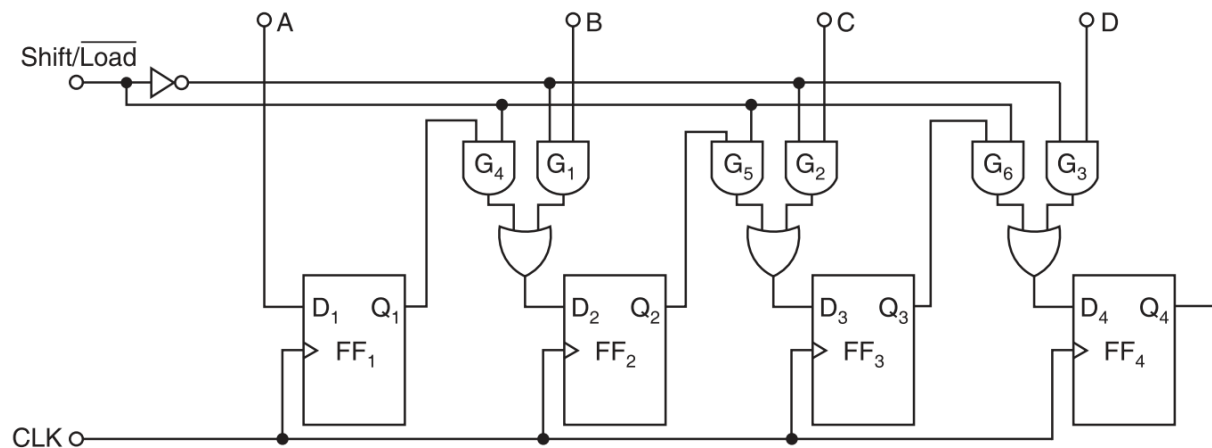
$$M_0 \bar{Q}_0 + \bar{M}_0 Q_0 = T_1$$

1. State the clear difference between a register and a counter. [1 Marks]

Solution: A register is a general-purpose storage unit that holds data, while a counter is a specific type of register that changes its value (increments or decrements) based on a clock signal to count events.

2. Draw and explain the circuit diagram of a Parallel-In Serial-Out (PISO) shift register, highlighting the operation under shift and load conditions. [4 Marks]

Solution:



Explanation:

- ❖ For a parallel-in, serial-out, shift register, the data bits are entered simultaneously into their respective stages on parallel lines, but the data bits are transferred out of the register serially, i.e. on a bit-by-bit basis over a single line.
- ❖ There are four data lines A, B, C, and D through which the data is entered into the register in parallel form.
- ❖ The signal Shift/ $\overline{LOAD}$ allows (a) the data to be entered in parallel form into the register and (b) the data to be shifted out serially from terminal Q₄
- ❖ When Shift/ $\overline{LOAD}$ line is HIGH, gates G₁, G₂, and G₃ are disabled, but gates G₄, G₅ and G₆ are enabled allowing the data bits to shift-right from one stage to the next.
- ❖ When Shift/ $\overline{LOAD}$ line is LOW, gates G₄, G₅, and G₆ are disabled, but gates G₁, G₂ and G₃ are enabled allowing the data input to appear at the D inputs of the respective FFs.
- ❖ When a clock pulse is applied, these data bits are shifted to the Q output terminals of the FFs and, therefore, data is inputted in one step. The OR gate allows either the normal shifting operation or the parallel data entry depending on which AND gates are enabled by the level on the Shift/ $\overline{LOAD}$ input.

3. Design a 4-bit Gray to Binary code converter with proper derivation and corresponding logic diagram. [5 Marks]

Solution:

4-bit Gray				4-bit binary			
G_4	G_3	G_2	G_1	B_4	B_3	B_2	B_1
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	1	0	0	1	0
0	0	1	0	0	0	1	1
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
0	1	0	1	0	1	1	0
0	1	0	0	0	1	1	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	1	1	0	1	0
1	1	1	0	1	0	1	1
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1
1	0	0	1	1	1	1	0
1	0	0	0	1	1	1	1

$$B_4 = \Sigma m(12, 13, 15, 14, 10, 11, 9, 8) = \Sigma m(8, 9, 10, 11, 12, 13, 14, 15)$$

$$B_3 = \Sigma m(6, 7, 5, 4, 10, 11, 9, 8) = \Sigma m(4, 5, 6, 7, 8, 9, 10, 11)$$

$$B_2 = \Sigma m(3, 2, 5, 4, 15, 14, 9, 8) = \Sigma m(2, 3, 4, 5, 8, 9, 14, 15)$$

$$B_1 = \Sigma m(1, 2, 7, 4, 13, 14, 11, 8) = \Sigma m(1, 2, 4, 7, 8, 11, 13, 14)$$

G_2G_1	G_4G_3			
	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$$B_4 = G_4$$

K-map for B_4

G_2G_1	G_4G_3			
	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$$B_3 = G_4 \oplus G_3$$

K-map for B_3

$$B_3 = \bar{G}_4 G_3 + G_4 \bar{G}_3 = G_4 \oplus G_3$$

G_2G_1	G_4G_3			
	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$$B_2 = G_4 \oplus G_3 \oplus G_2$$

K-map for B_2

$$B_2 = \bar{G}_4 G_3 \bar{G}_2 + \bar{G}_4 \bar{G}_3 G_2 + G_4 \bar{G}_3 \bar{G}_2 + G_4 G_3 G_2$$

$$= \bar{G}_4 (G_3 \oplus G_2) + G_4 (\bar{G}_3 \oplus \bar{G}_2) = G_4 \oplus G_3 \oplus G_2 = B_3 \oplus G_2$$

		G_2G_1			
		00	01	11	10
G_4G_3	00	0	1	3	2
	01	4	5	7	6
	11	12	13	15	14
	10	8	9	11	10

$$B_1 = G_4 \oplus G_3 \oplus G_2 \oplus G_1$$

K-map for B_1

mans

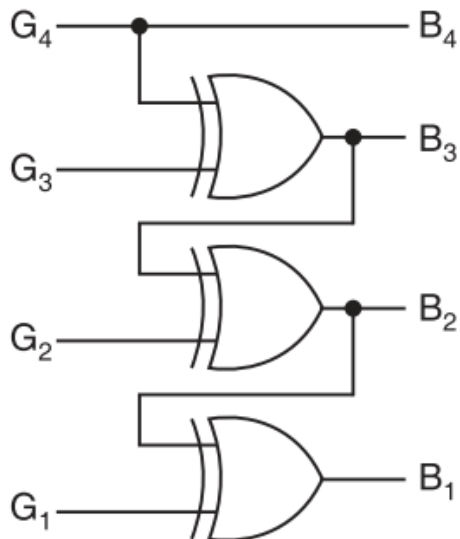
$$B_1 = \bar{G}_4\bar{G}_3\bar{G}_2G_1 + \bar{G}_4\bar{G}_3G_2\bar{G}_1 + \bar{G}_4G_3G_2G_1 + \bar{G}_4G_3\bar{G}_2\bar{G}_1 + G_4G_3\bar{G}_2G_1 + G_4G_3G_2\bar{G}_1 + G_4\bar{G}_3\bar{G}_2\bar{G}_1 + G_4\bar{G}_3G_2G_1$$

$$= \bar{G}_4\bar{G}_3(G_2 \oplus G_1) + G_4G_3(G_2 \oplus G_1) + \bar{G}_4G_3(\overline{G_2 \oplus G_1}) + G_4\bar{G}_3(\overline{G_2 \oplus G_1})$$

$$= (G_2 \oplus G_1)(\overline{G_4 \oplus G_3}) + (G_2 \oplus G_1)(G_4 \oplus G_3)$$

$$= G_4 \oplus G_3 \oplus G_2 \oplus G_1$$

$$= B_2 \oplus G_1$$



Question 5

a) Write the function of the following pins of 8085 microprocessor
 ALE: Address Latch Enable. This pin is used to demultiplex address and data bus. If ALE = 1, AD0-AD7 & act as address bus; if ALE = 0, AD0-AD7 act as data bus.

Reset Out: This pin is used to reset the peripheral devices connected to the microprocessor based system. When processor is reset by Reset in pin, processor also reset the other peripherals.

HLDA: Hold acknowledge; used for direct memory access (DMA) operation,

SOD: Serial output data. This pin is used to send the data from microprocessor to output device serially.

b) After the execution of the following program, write the content of the Accumulator, memory 5590H, 5591H in hexadecimal number system. Also write the content of the carry, Auxiliary carry, and zero flag in binary form.

$LXI\ H\ 5590\ H \rightarrow [HL] = 5590\ H$
 $MVI\ A\ F0\ H \rightarrow [A] = F0\ H$
 $MOV\ M,\ A \rightarrow [5590] = F0\ H$
 $MVI\ A\ 0F\ H \rightarrow [A] = 0F\ H$
 $ADD\ M \rightarrow [A] = [A] + [M] = 0F + F0 = FF\ H$
 $INR\ A \rightarrow [A] = [A] + 01 \Rightarrow FF + 1 = 1[00]_{16}$
 $STA\ 5591\ H \rightarrow [5591] = [A] = 00$
 HLT

$\therefore [A] = 00\ H$
 $[5590] = F0$
 $[5591] = 00$
 $CY = 1$
 $AC = 0$
 $Z = 1$

Handwritten notes:
 INR inst. does not change the carry bit. $\therefore CY = 0$ has been
 * marks given for both $CY = 0$ & $CY = 1$

c) Write the content of memory locations 4FFF H and 5000 H (in hex code) after the execution of the following program.

$MVI\ B\ 23\ H \rightarrow [B] = 23\ H$
 $MVI\ C\ 34\ H \rightarrow [C] = 34\ H$
 $LXI\ H\ 4332\ H \rightarrow [HL] = 4332\ H$
 $DAD\ B \rightarrow [BC] + [HL] = 2334 + 4332 = 6666 = [66][66]$
 $MOV\ A,\ H \rightarrow [A] = [H] = 66$
 $STA\ 5000\ H \rightarrow [5000] = [A] = 66$
 $MOV\ A,\ L \rightarrow [A] = [L] = 66$
 $STA\ 4FFF\ H \rightarrow [4FFF] = [A] = 66$
 HLT

$[5000] = [H] = 66\ H$
 $[4FFF] = [L] = 66\ H$

d) Write the content of memory location 2500H, registers A and B in hex code, after the execution of the following program.

$XRA\ A \rightarrow A \oplus A = 00\ H$
 $MOV\ B,\ A \rightarrow [B] = 00\ H$
 $INR\ B \rightarrow [B] = [B] + 1 = 00 + 1 = 01\ H$
 $ORI\ 0F\ H \rightarrow [A] = [A] OR 0F = 0F\ H$
 $ADD\ B \rightarrow [A] = [A] + [B] = 0F + 01 = 10\ H$
 $STA\ 2500\ H \rightarrow [2500] = [A] = 10\ H$
 $CMA \rightarrow [A] = \text{complement of } A = EF\ H$
 HLT

$[A] = 0F\ H$
 $[2500] = 10\ H$
 $[A] = EF\ H$
 $[B] = 01\ H$

$\therefore [2500] = 10\ H$
 $[A] = EF\ H$
 $[B] = 01\ H$